Solutions of Assignment #1 part 1, Total points: 40
(Course: CS 401)

These are the first two problems for Assignment 1. **The remaining problems of Assignment 1 will be given out later.**

Problem 1 (20 points): Prof. Smart thinks he is smarter than all the students in this CS 401 class. He has made the following claim to show how smart he is.

> **Claim made by Prof. Smart**: Consider the stable matching problem as taught in class. Suppose that we have only three men, say m_1, m_2 and m_3, and only three women, say w_1, w_2 and w_3, with their corresponding preference lists. Suppose also that the matching m_1–w_1, m_2–w_2, m_3–w_3 is a stable matching. Prof. Smart claims that in that case the matching m_1–w_2, m_2–w_1, m_3–w_3 can **never** be a stable matching.

Your task is to decide if Prof. Smart is indeed so smart. For this purpose, do the following.

- Either prove the claim made by Prof. Smart is indeed correct. Such a proof should work **no matter** what the preferences of the men and women are, as long as m_1–w_1, m_2–w_2, m_3–w_3 is a stable matching.
- Or, prove the claim made by Prof. Smart is wrong by giving a counter-example. The counter-example should provide the preferences lists of every man and woman, and show that for these preference lists **both** m_1–w_1, m_2–w_2, m_3–w_3 and m_1–w_2, m_2–w_1, m_3–w_3 are indeed stable matchings.

Solution: The following preference lists constitute a counter-example since both both m_1–w_1, m_2–w_2, m_3–w_3 and m_1–w_2, m_2–w_1, m_3–w_3 are stable matchings.

	first	second	third
m_1	w_1	w_2	w_3
m_2	w_2	w_1	w_3
m_3	w_3	w_1	w_2

	first	second	third
w_1	m_2	m_1	m_3
w_2	m_1	m_2	m_3
w_3	m_3	m_1	m_2

Problem 2 (20 points): Take the following functions and arrange them in ascending order of growth rates. That is, if function $g(n)$ immediately follows function $f(n)$ in your list, then it should be the case that $f(n) = O(g(n))$. **Briefly justify your answer**.

$$g_1(n) = 3^{\sqrt{\log_2 n}}$$
$$g_2(n) = 2^n$$
$$g_3(n) = n^{300}$$
$$g_4(n) = n(\log_2 n)^{100}$$
$$g_5(n) = \frac{n^{\log_2 n}}{2345}$$
$$g_6(n) = 2^{2^n}$$
$$g_7(n) = 2^{n^2}$$

The order is $g_1(n)$, $g_4(n)$, $g_3(n)$, $g_5(n)$, $g_2(n)$, $g_7(n)$, $g_6(n)$, *i.e.*,

$$3^{\sqrt{\log_2 n}}, \quad n(\log_2 n)^{100}, \quad n^{300}, \quad \frac{n^{\log_2 n}}{2345}, \quad 2^n, \quad 2^{n^2}, \quad 2^{2^n}$$

To simplify notations, lets us use the symbol $\prec$ in the following manner: $f \preceq g$ means $f = O(g)$.

- As observed in class,

$$d \preceq \log_c n \preceq n^b \preceq a^n \text{ for any constant } a, b, c, d > 1$$

- So we have

$$n \preceq n^2 \preceq 2^n \text{ which implies } 2^n \preceq 2^{n^2} \preceq 2^{2^n}$$

- and $\frac{n^{\log_2 n}}{2345} \preceq 2^n$ since by taking logarithm of both sides (base 2) we get $(\log_2 n)^2 - \log_2 2345 \preceq n$.

- and

$$300 \preceq \log_2 n \text{ which implies } n^{300} \preceq n^{\log_2 n} \text{ which implies } n^{300} \preceq \frac{n^{\log_2 n}}{2345}$$

- and

$$\sqrt{n} = n^{\frac{1}{2}} \preceq 300n \text{ which implies } (\log_2 n)^{\frac{1}{2}} \preceq 300 \log_2 n$$
$$\text{which implies } 2^{\sqrt{\log_2 n}} \preceq 2^{300 \log_2 n} = n^{300}$$

- and

$2^{\sqrt{n}} \preceq 2^n n^{100}$ which implies $2^{\sqrt{\log_2 n}} \preceq 2^{\log_2 n} (\log_2 n)^{100}$ which implies $2^{\sqrt{\log_2 n}} \preceq n(\log_2 n)^{100}$

- and $n(\log_2 n)^{100} \preceq n^{300}$ since by taking logarithm of both sides (base 2) we get $100 \log_2 n + \log_2 \log_2 n \preceq 300 \log_2 n$, which is true.